ÖAW AUSTRIAN ACADEMY OF SCIENCES

austrian centre for digital humanities & cultural heritage

Intro to CBS4DH

Dimitra Grigoriou
2026-11-11
136143 KU Computational Background Skills for Digital Humanities (2026W) (Group B)

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Dimitra Grigoriou

- Background in Linguistics, English Lit.
- Since August 2024 at IHB
- Data modeling and Technical Support
- Currently:
  - TEI-C Board Member
  - sources on Habsburg-Ottoman diplomacy QHOD
  - Minutes of the Councils of Ministers

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Index

- How this course works
- Get to know each other
- CBS4DH: what does that mean?
- Three values in DH
  - Collaboration
  - Openness
  - Versatility

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How this course works

- Six parts
  - File formats and encodings (**11.11.**)
  - Command line (**12.11.**)
  - Minimal introduction to web technologies (**18.11.**)
  - Regular expressions (**19.11.**)
  - Git (**25.11.**)
  - Tools and resources for Digital Humanities (**26.11.**)
- Course material: https://github.com/acdh-oeaw/Teaching_CBS4DH/

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How this course works

- **Attendance** is extremely important! **Absence:** 1,5 Modules
- Complete the **exercises** on **Moodle. Deadline:** Tuesday Dec. 1st. 2026, 09:00 am
- Please **follow along** with the tutorials
- **Ask questions**
- **Check** if you have installed everything
  - Visual Studio Code: https://code.visualstudio.com/
  - (only Windows users) Gitbash: https://gitforwindows.org/
  - Create a GitHub account: https://github.com/join

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Two questions for you

- What are you studying / have you studied previously?
- What operating system do you use?

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What is Digital Humanities?

- The application of digital methods to research in the Humanities
- Varying degree of use of digital methods
- https://whatisdigitalhumanities.com



- Around 800 definitions of DH
- Raw data

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What Is Digital Humanities?

New quote

The best way I've found to describe what Digital Humanities is, in first place to myself, is using a quote from Thomas Pynchon's Crying of Lot 49: 'You know what a miracle is ... another world's intrusion into this one' **Federico Meschini, De Montfort University**

NB: Click the New quote button to display a new random quote. Quotes were pulled from participants from the Day of DH between 2009-2014. As of January 2015, the database contains 817 rows and randomly selects a quote each time the page is loaded. If you want to do something cool with the data, I am providing the code and data here.

Made by Jason Heppler. Problems? Questions? I'm @jaheppler on Twitter.

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Not only Humanities

- Social Sciences and Humanities (**SSH**)
  - For example, SSHOC (Social Sciences and Humanities Open Cloud/Cluster)
- Political science
- Business
- Broad application of certain techniques/research areas
  - e.g. Natural Language Processing
- And many others

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„Push your changes to the repo“

CSS

```
^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$
```

HTML

„You can write a shell script to do this“

131.130.70.8

```
Alice,25,New York,Engineer
Bob,30,Los Angeles,Doctor
Charlie,35,Chicago,Artist
```

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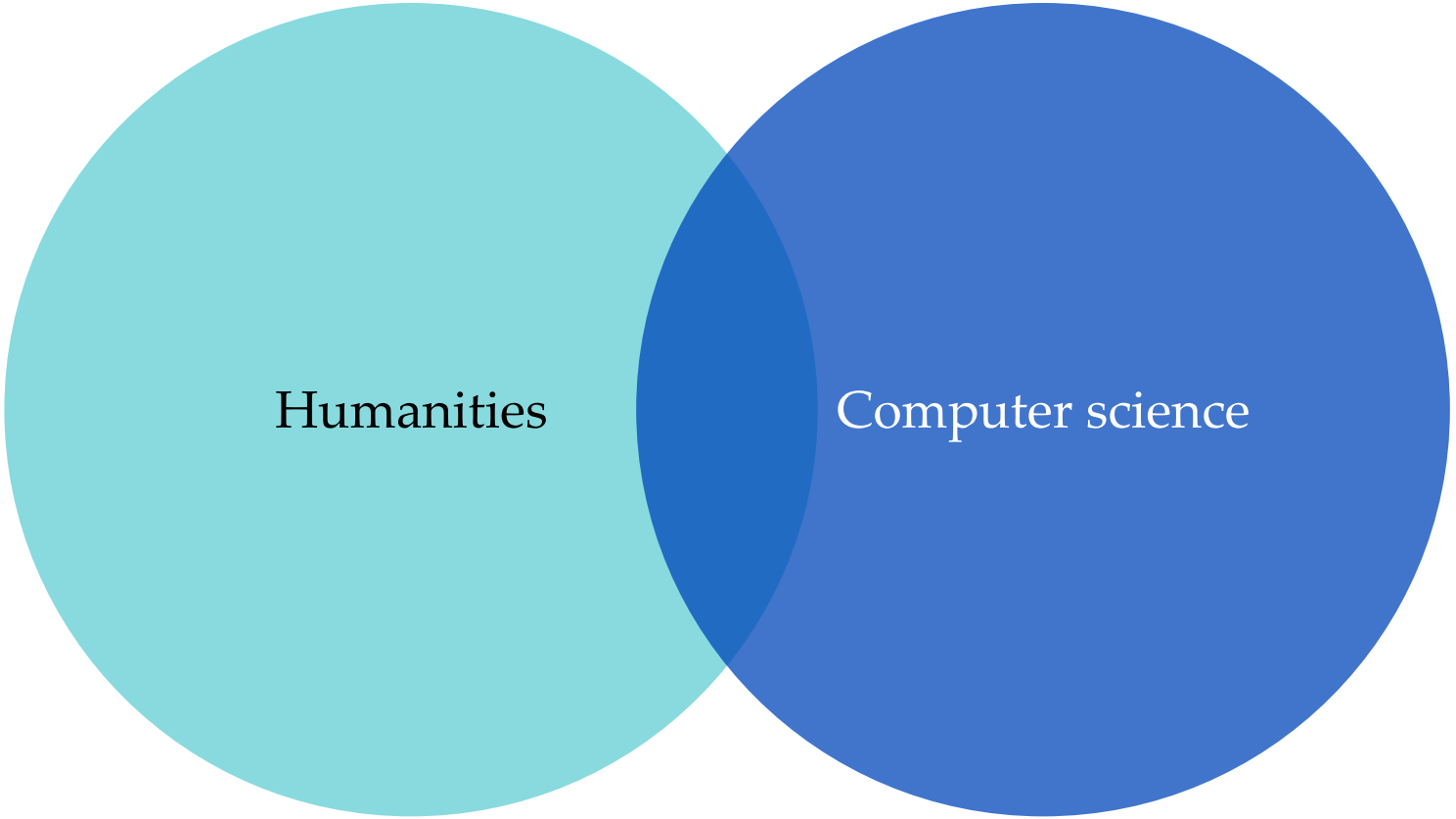
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Some things you can do with these skills

- Print a list of all files in a directory
- Find all instances of “colour” and “color” with a single search
- Keep a change history of your document
- Create a basic website
- Find new opportunities to learn
- ... and many more!

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Three ‚values'
in DH

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Collaboration

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Communities and collaboration

- DH project = no one-man show!
- Collaboration between different actors, stakeholders and communities
  - Digital Humanists
  - Developers
  - More “traditional” scholars
  - Different departments / institutions
  - Broader audience



One Man Band, CC BY 2.0, Created by Jarekt

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Digital Humanities Centers / Institutes in Austria

- Digital Humanities @ University of Vienna
- Austrian Centre for Digital Humanities (at ÖAW)
- University of Graz – Institut für digitale Geisteswissenschaften
- Lists
  - EADH Digital Humanities Centers
  - NYU List of Digital Humanities Centers

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Austrian Centre for Digital Humanities (ACDH)

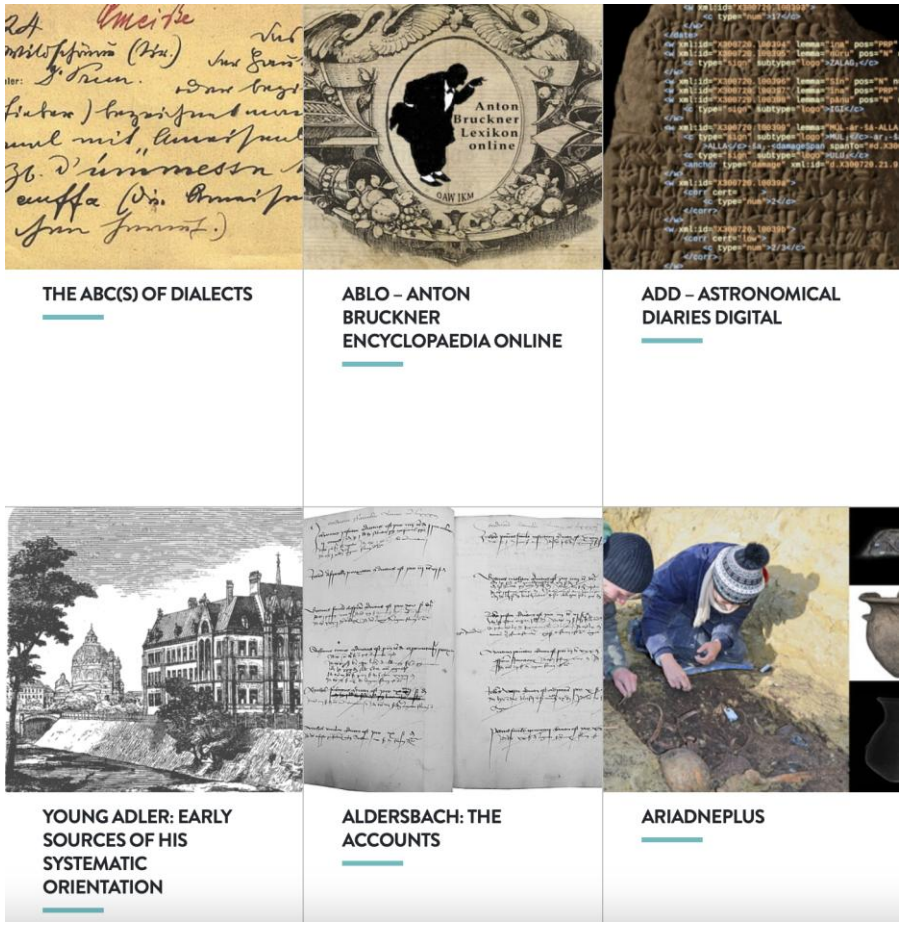
- Founded in 2015, concept from 2013
- Mission: *fostering the use of digital methods and tools in the humanities research in Austria*
- Rapid growth
  - From 12 in 2015 to 141 in 2024
  - ~95 projects



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Austrian Centre for Digital Humanities (ACDH)

- Four research units
  - Linguistics
  - Literature and Text
  - Musicology
  - DHRI – Digital Humanities Research & Infrastructure
- Both a research *and* a service-providing institution (What we offer)



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DH associations

- DHd – Digital Humanities im deutschsprachigen Raum https://digitalhumanities.de
- EADH – European Association for Digital Humanities http://eadh.org
- ADHO – Alliance of Digital Humanities Organizations https://adho.org (list of partners: https://adho.org/membership-individual/)
- MLA2S – Machine Learning at the Austrian Academy of Sciences http://www.oeaw.ac.at/mla2s

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Research Infrastructures

- "facilities that provide resources and services for research communities to conduct research and foster innovation" (European Commission)
  - No humans mentioned!
- Networks of people and expertise
- Both bottom-up and top-down
  - Exchange of ideas and knowledge sharing
  - Consolidation, harmonization (standards, best practices)
  - Provision of services

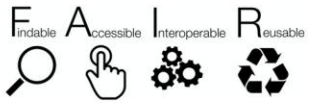
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The European Research Landscape

Role: Interoperable Data Aggregation (beyond metadata)

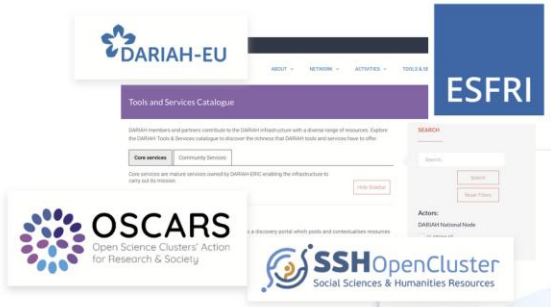






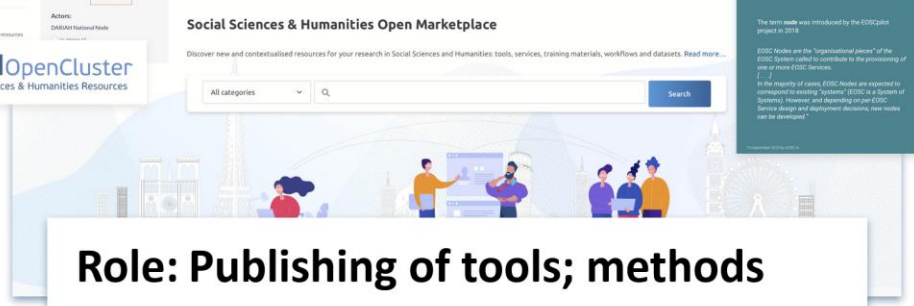
Role: State of the Art processing of Cultural Heritage Data (e.g AI-based)













Role: Publishing of tools; methods and derived datasets for reuse

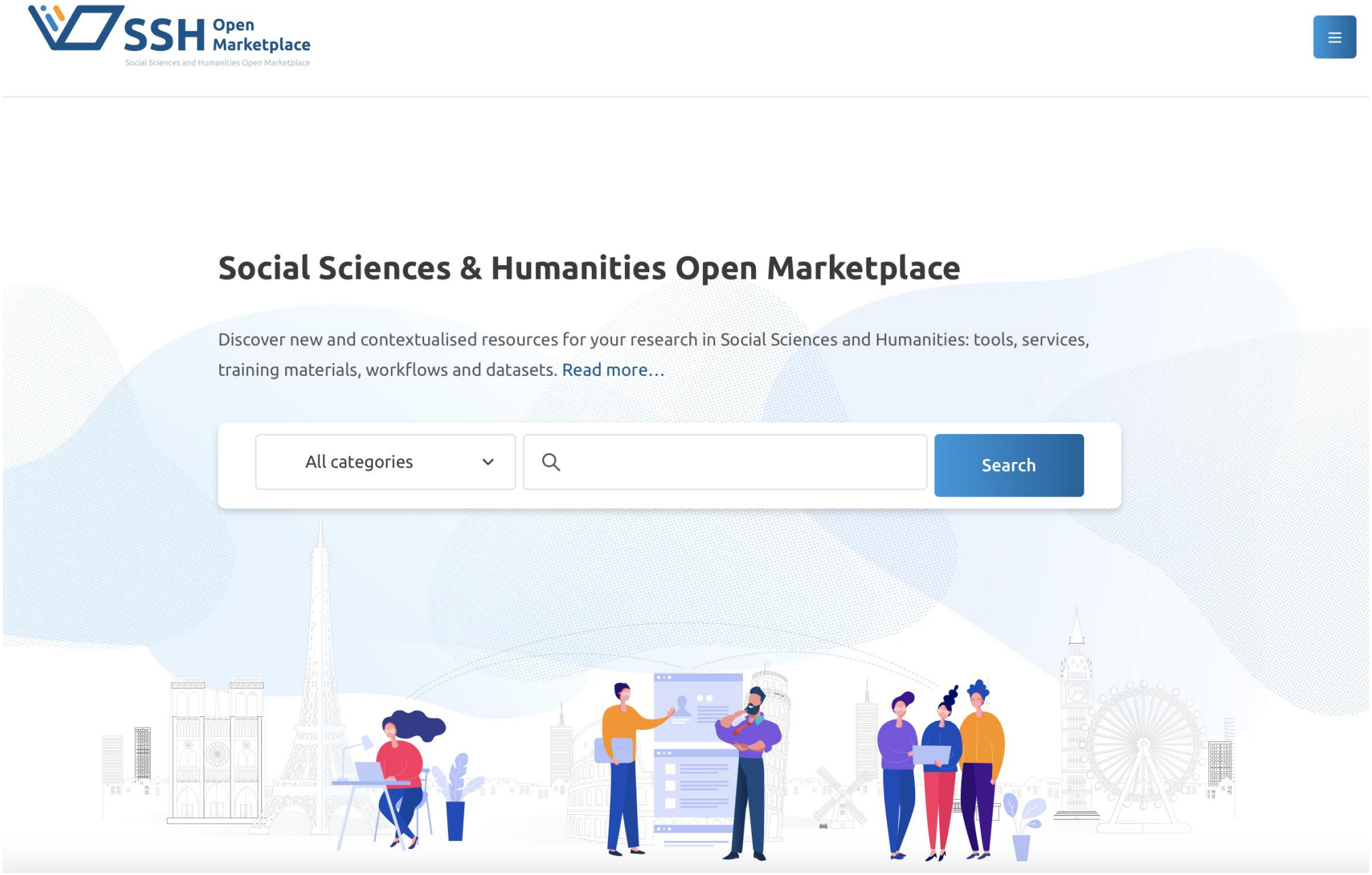
By Sally Chambers

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DARIAH-EU

DARIAH

- European *Digital Research Infrastructure for the Arts and Humanities*
- Strong voice for the DH community on the European level
- Lines of action
  - Common policies & guidelines
  - Dissemination and discoverability of resources (SSH Open Marketplace)
  - Knowledge sharing (DARIAH Campus)
  - Integration with other initiatives, e.g. European Open Science Cloud (EOSC)



[SSH Open Marketplace]

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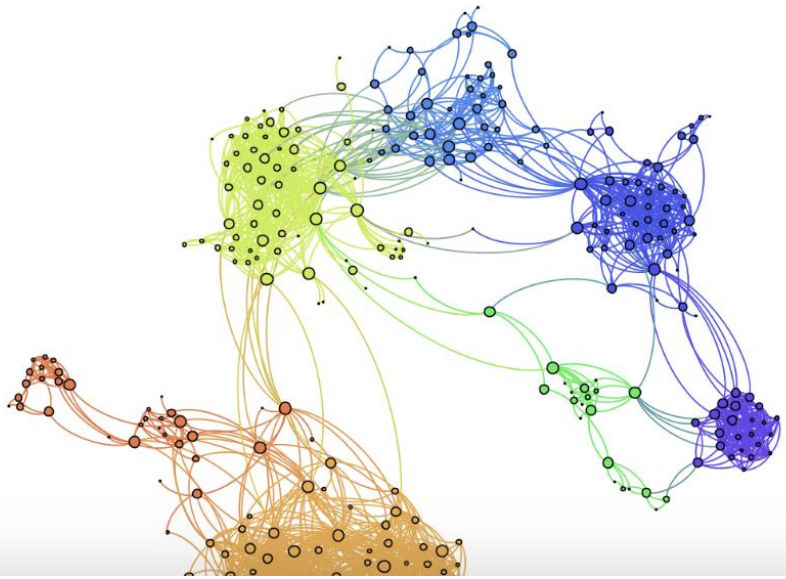
Gephi

Copy to clipboard

Gephi is the leading visualization and exploration software for all kinds of graphs and networks.

Gephi is open-source and free. Runs on Windows, Mac OS X and Linux.

Go to Tool or service



Details

ACCESS

License Common Development and Distribution License 1.0 GNU General Public License v3.0 only

Terms Of Use Free

CATEGORISATION

Activity Capturing Modeling Network Analysis Analyzing Visual Analysis Data Visualization Relational Analysis Spatial Analysis Content Analysis Discovering

Keyword Models graphs french R faceted browser graph streaming recommended

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CLARIN
Common Language Resources and Technology Infrastructure

CLARIN

- *Common Language Resources and Technology Infrastructure*
- Easy and sustainable access to a broad range of language data and tools
- Distributed research infrastructure, with centers all over Europe
- Services
  - Content search (search for specific patterns across collections)
  - Virtual Language Observatory (explore language resources and tools)
  - Language Resource Switchboard (process your data with specialized tools)
  - Learning Hub and DH Course Registry (find learning opportunities in DH)

DIGITAL HUMANITIES COURSE REGISTRY

CLARIN

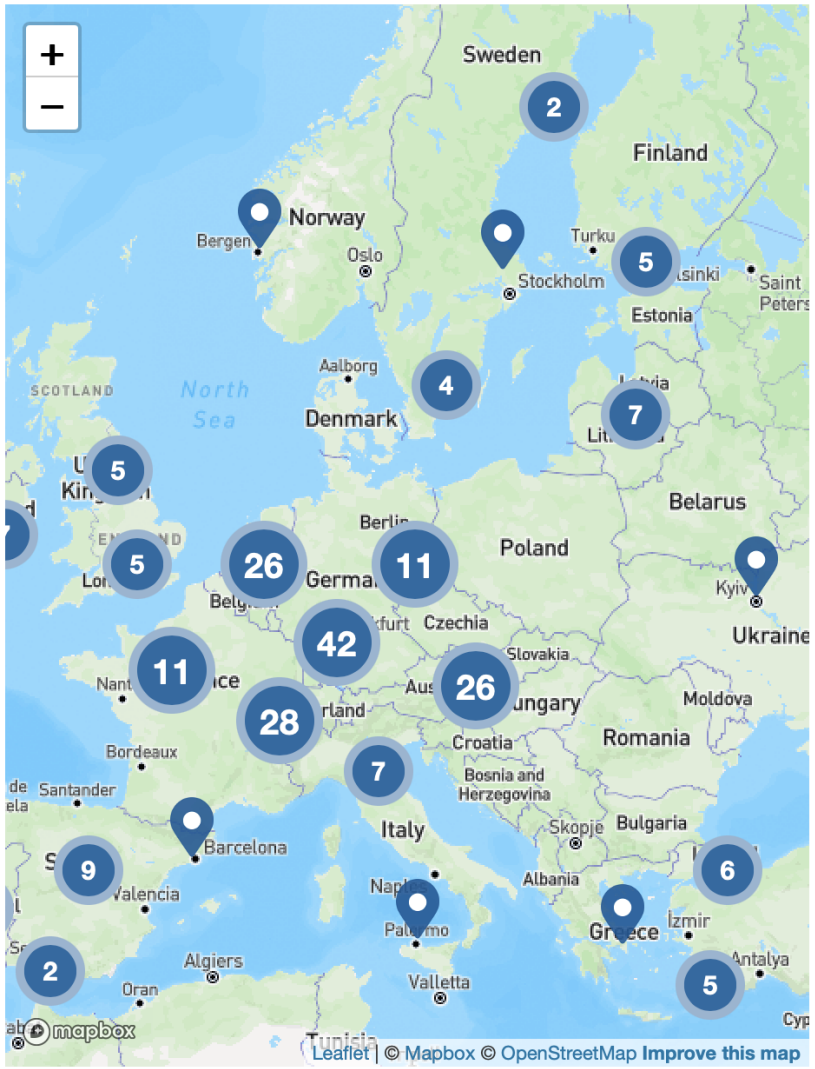
DARIAH-EU
Digital Research Infrastructure for the Arts and Humanities

INFO

FILTER SORT

Find a course directly by name or university

NAME	UNIVERSITY	PLACE	DATE	TYPE
MA in Digital Humanities	King's College London	London, United Kingdom	24 Sept 1 years	Master Programme
MSc in Digital Scholarship	University of Oxford	Oxford, United Kingdom	9 Oct 11 months	Master Programme
Digital Stylometry	Freie Universität Berlin	Berlin, Germany	17 Apr 2025 1 semester	Course
Introduction to the Digital Humanities	Freie Universität Berlin	Berlin, Germany	14 Apr 1 semester	Course
Postgraduate Certificate in Digital Humanities and Culture	Trinity College Dublin	Dublin, Ireland	5 Sept 2025 2 semester	Course
Postgraduate Diploma in Digital Humanities & Culture	Trinity College Dublin	Dublin, Ireland	5 Sept 2 semester	Course
MPhil in Digital Humanities and Culture	Trinity College Dublin	Dublin, Ireland	5 Sept 1 years	Master Programme
Swedish National Research School in Digital Humanities: Dat ...	Uppsala University	Uppsala, Sweden	25 Sept 12 months	PhD Programme
Digital Humanities	Higher School of Economics Perm	Perm, Russian Federation	1 Sept 2 years	Master Programme
Language and computers	Universitetet i Bergen	Bergen, Norway	15 Jan 1 semester	Bachelor Programme
Master in Textual Heritage & Digital Humanities [Máster en ...	Universidad de Salamanca	Salamanca, Spain	23 Sept 2 semester	Master Programme
Corpus-Driven Lexicology	Mykolas Romeris University	Vilnius, Lithuania	1 Feb 5 months	Bachelor Programme



Imprint FAQ

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DH Course Registry

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CLARIAH-AT

Austrian DH landscape

- CLARIAH-AT – clariah.at
  - CLARIN + DARIAH at the Austrian level
- DiTAH – ditah.at
  - *Digitale Transformation der österreichischen Geisteswissenschaften*
  - Four-year project (2020-2024) for the establishment of Digital Humanities in Austria
- Funding schemes (e.g. CLARIAH-AT Project funding, Funding for Early Career Researchers, Transkribus Scholarship)
- Next DHd 2027 Conference will take place in Marburg!
  (1st – 5th March 2027)

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CLARIAH-AT Partners

CLARIAH-AT

ÖAW AUSTRIAN ACADEMY OF SCIENCES

Österreichische Nationalbibliothek

nhm naturhistorisches museum wien

University for Continuing Education Krems

DONAU UNIVERSITÄT KREMS

UNIVERSITÄT GRAZ
UNIVERSITY OF GRAZ

UNI GRAZ

mdw university of music and performing arts vienna

universität innsbruck

UNIVERSITÄT KLAGENFURT

PARIS LODRON UNIVERSITÄT SALZBURG

universität wien

CEU CENTRAL EUROPEAN UNIVERSITY

acdh

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Cultural Heritage (CH) Infrastructures

- Improve the availability of CH objects for research and beyond
- Particularly relevant for GLAM sector (Galleries, Libraries, Archives and Museums)
- Austrian level: Kulturpool
- European level: Europeana
- ECCCH
  (European Collaborative Cloud for Cultural Heritage)



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Kulturpool

Austrian CH aggregator

europeana

European CH aggregator

Slovakiana

Swedish Open Cultural Heritage

culturaitalia

Culturaltalia

... and many others

[Full list of Europeana aggregators]

National Heritage Institute, Bucharest

Greek Aggregator SearchCulture.gr

BnF Gallica

Gallica

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Openness

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Website
Book/Article
Digital edition

Below the surface...

Data

Scans
Photos
Database
3D models
Transcriptions
Audio recordings
Video recordings
Reports and notes
...



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Data is/are important

- Data is the backbone of your research, even in the Humanities
- Open data allows you to validate your results
- Open data allows others to build on your research
- **Data first!** Research data as one of the most important outputs of projects
  - Datasets as first-class citizens in the research landscape
  - Can be published in repositories like Zenodo ...
  - ... and referenced as publications (e.g. with a DOI)



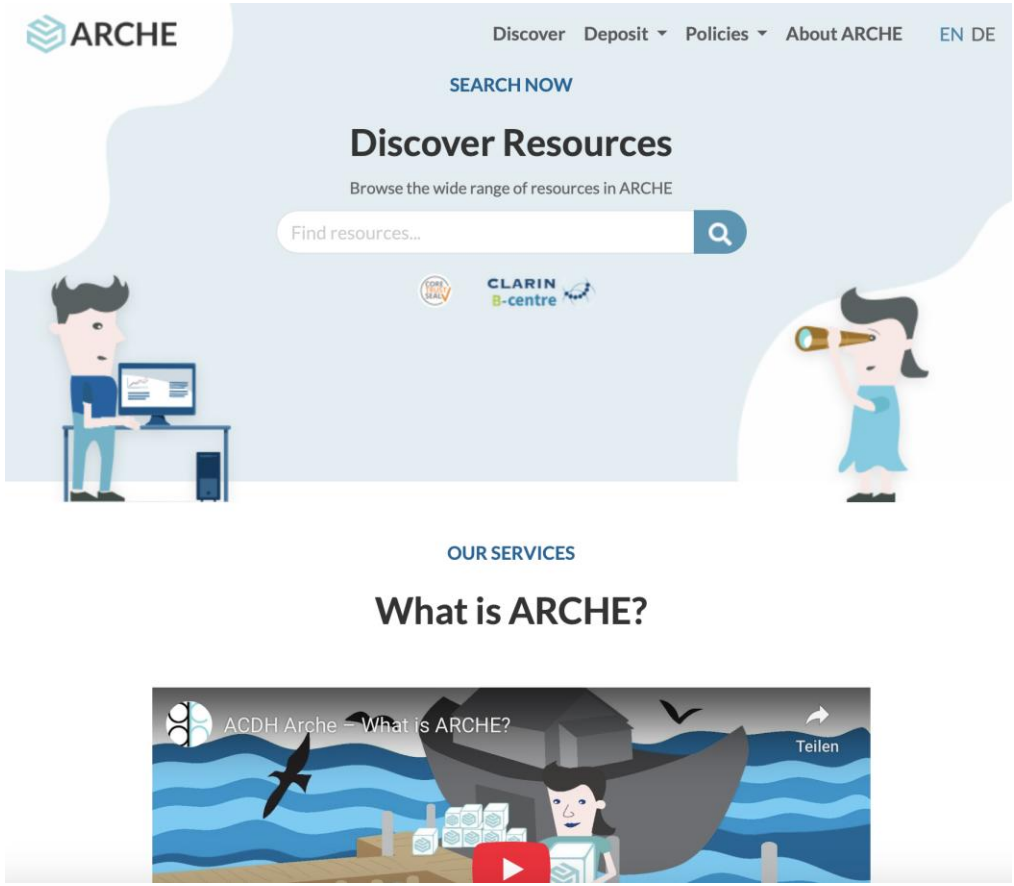


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ARCHE

ARCHE

- Repository for digital research data in the Humanities
- Developed and hosted by ACDH
- Curation team that helps in deposition process, checks & dissemination
- Integration with CLARIN VLO and other European catalogues



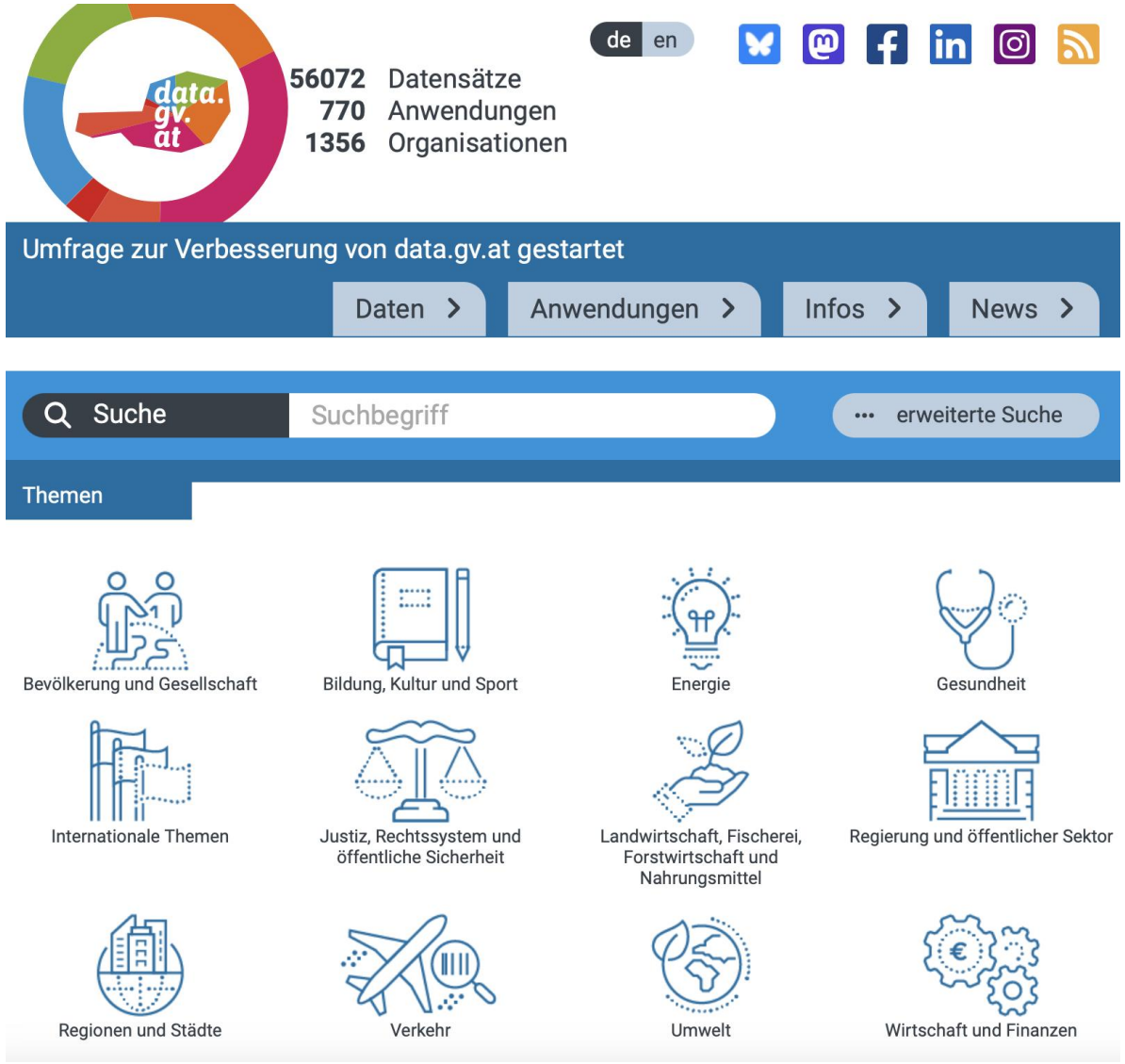
CORE TRUST SEAL

CLARIN B -centre

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Open Data

- FAIR Principles
  - Findability
  - Accessibility
  - Interoperability
  - Reusability
- Governments and local authorities
  - https://www.data.gov.uk
  - https://www.data.gv.at



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OpenAI

“Openness” in modern trends

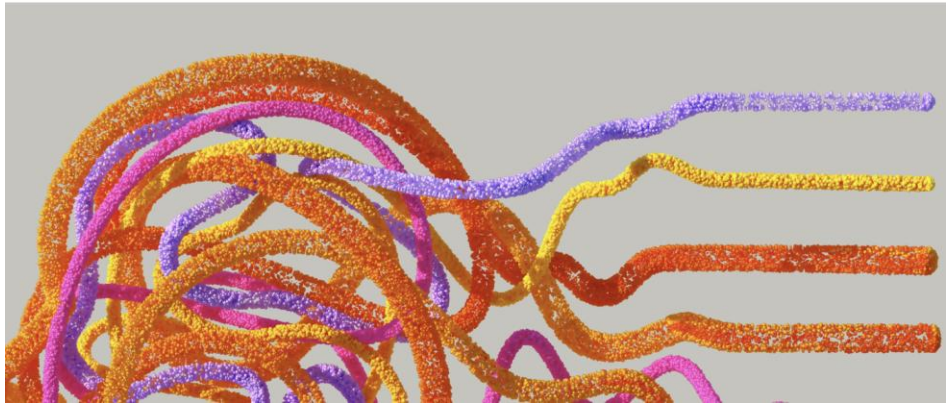
WORLD ECONOMIC FORUM

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EMERGING TECHNOLOGIES

What is open-source AI and how could DeepSeek change the industry?

Feb 5, 2025



Bundesministerium
Bildung, Wissenschaft
und Forschung

… › Strategien › Open Innovation Strategie

Open Innovation Strategie

Österreich ist der erste europäische Mitgliedsstaat, der eine umfassende nationale Open Innovation Strategie entwickelt hat.
Im Zuge eines breiten Beteiligungsprozesses, in welchem nicht nur Stakeholder aus Forschung, Wissenschaft und Wirtschaft sondern auch die Zivilbevölkerung eingebunden waren, wurde im Jahr 2016 eine Open Innovation Strategie für Österreich erstellt. Diese umfasst neben einer Analyse des nationalen und internationalen Status Quo zum Thema Open Innovation auch eine Vision für 2025 und konkrete Maßnahmen zur Umsetzung.

open data institute

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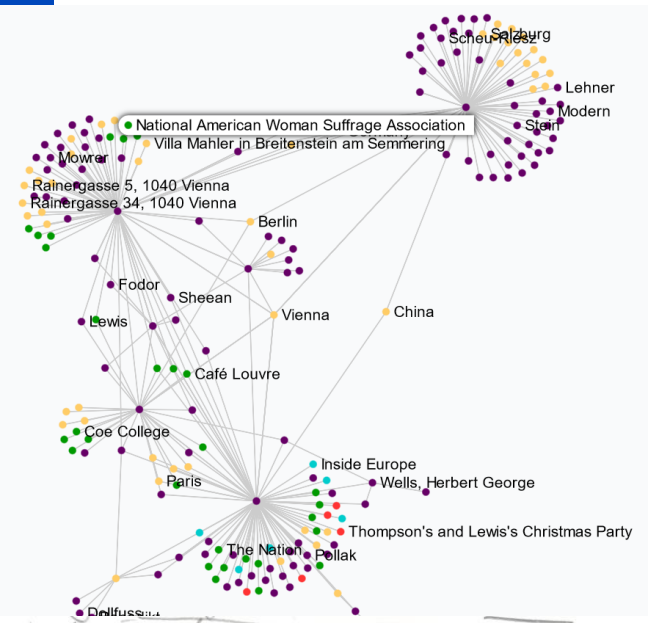
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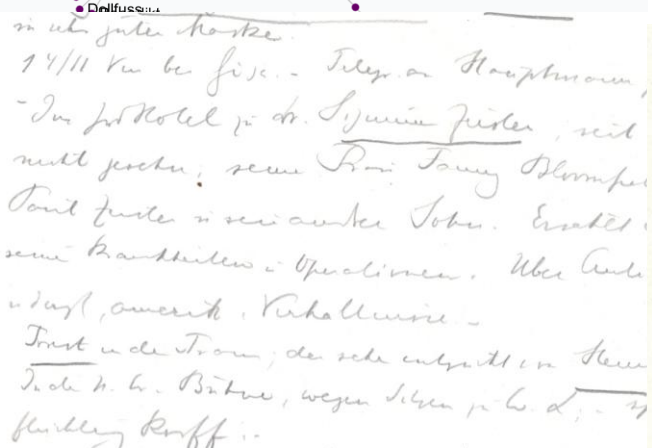
Versatility

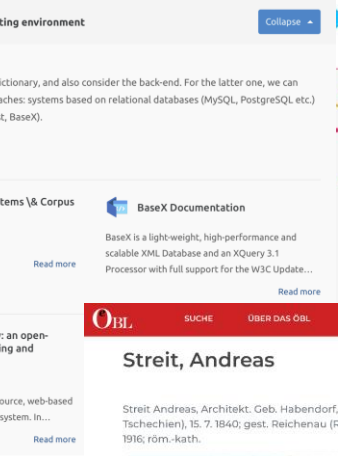
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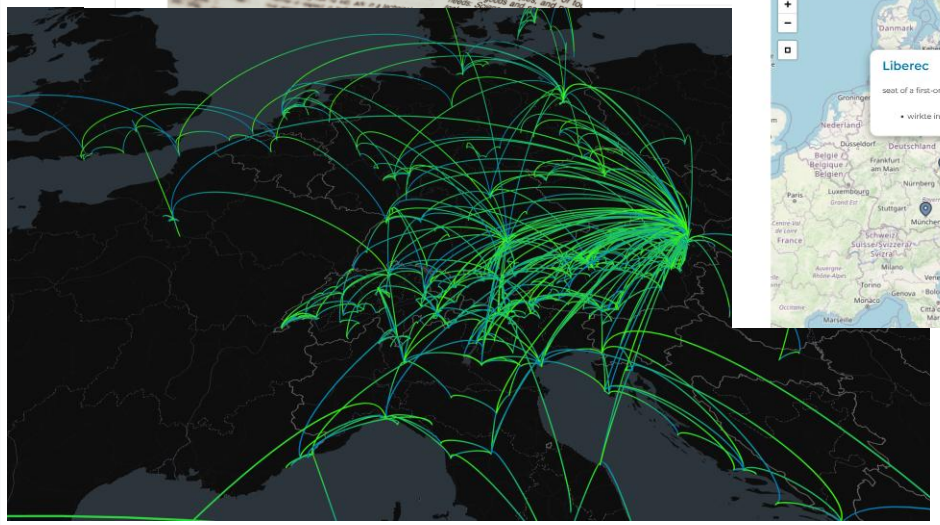
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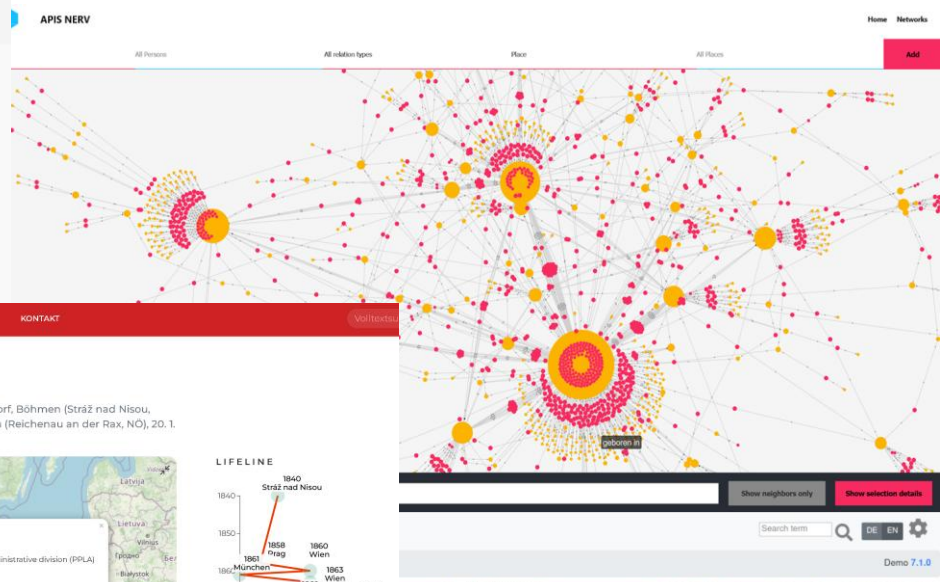
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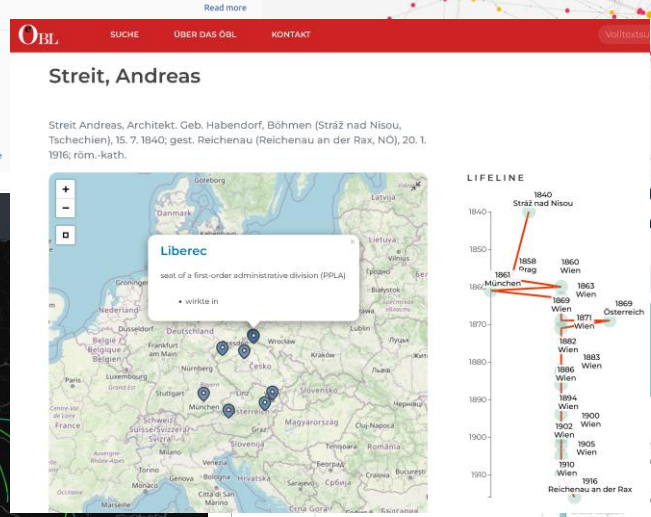


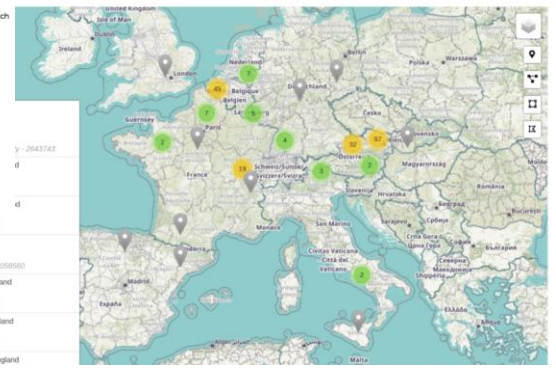


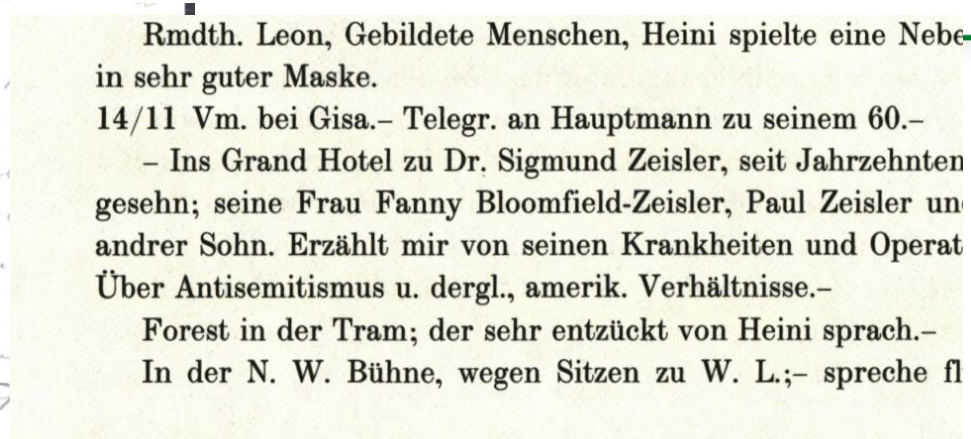












Rmdth. Leon, Gebildete Menschen, Heini spielte eine Nebe
in sehr guter Maske.
14/11 Vm. bei Gisa.– Telegr. an Hauptmann zu seinem 60.–
– Ins Grand Hotel zu Dr. Sigmund Zeisler, seit Jahrzehnten
gesehn; seine Frau Fanny Bloomfield-Zeisler, Paul Zeisler un
andrer Sohn. Erzählt mir von seinen Krankheiten und Operat
Über Antisemitismus u. dergl., amerik. Verhältnisse.–
Forest in der Tram; der sehr entzückt von Heini sprach.–
In der N. W. Bühne, wegen Sitzen zu W. L.;– spreche fl



Dienstag, 14. November 1922

14/11 Vm. bei Gisa.— Telegr. an Hauptmann zu seinem 60.—

— Ins Grand Hotel zu Dr. Sigmund Zeisler, seit Jahrzehnten nicht gesehn; seine Frau Fanny Bloomfield-Zeisler, Paul Zeisler und sein andrer Sohn. Erzählt mir von seinen Krankheiten und Operationen. Über Antisemitismus u. dergl., amerik. Verhältnisse.—

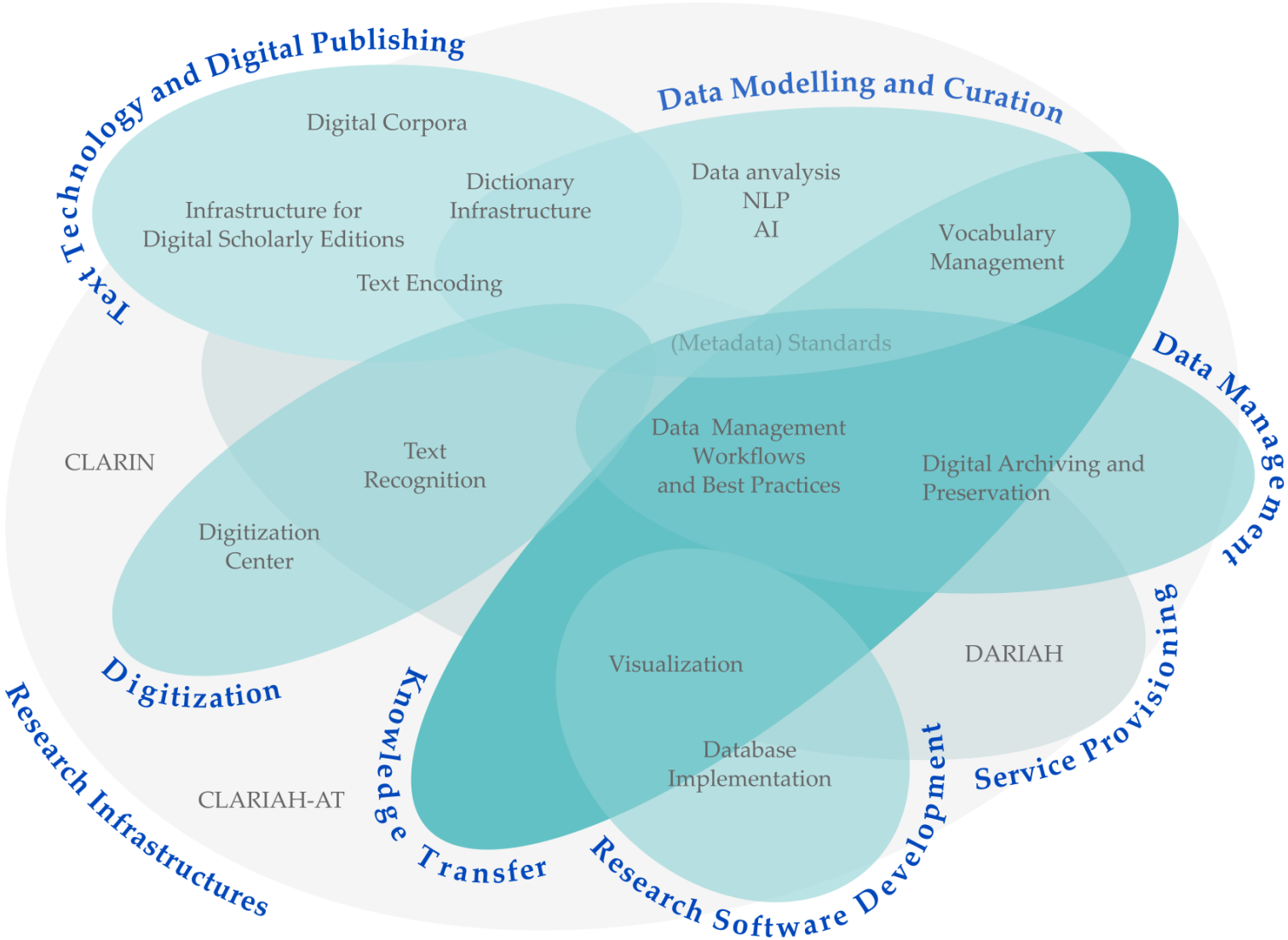
Forest in der Tram; der sehr entzückt von Heini sprach.—

In der N. W. Bühne, wegen Sitzen zu W. L.;— spreche flüchtig Korff.—

Mit Kolap Rmdth. „Weber“. Heini spielte den Candidaten und den zweiten jungen Weber. Gute Inszenirung von Martin. Das Stück wirkte wieder außerordentlich. Ein Gestalter

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Areas of Expertise



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Technology trends

Perceptron

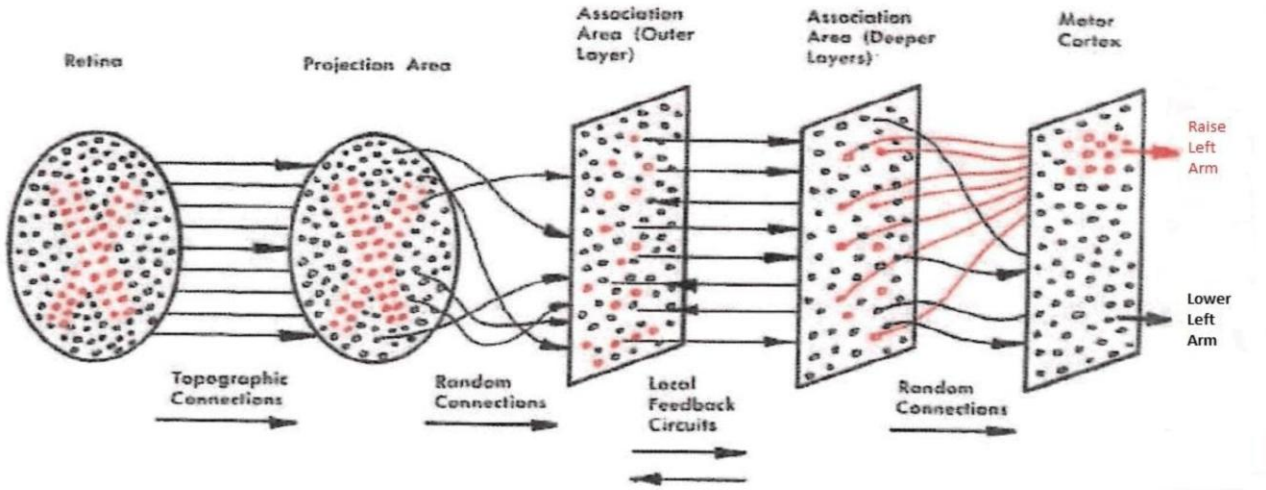


FIG. 1 — Organization of a biological brain. (Red areas indicate active cells, responding to the letter X.)

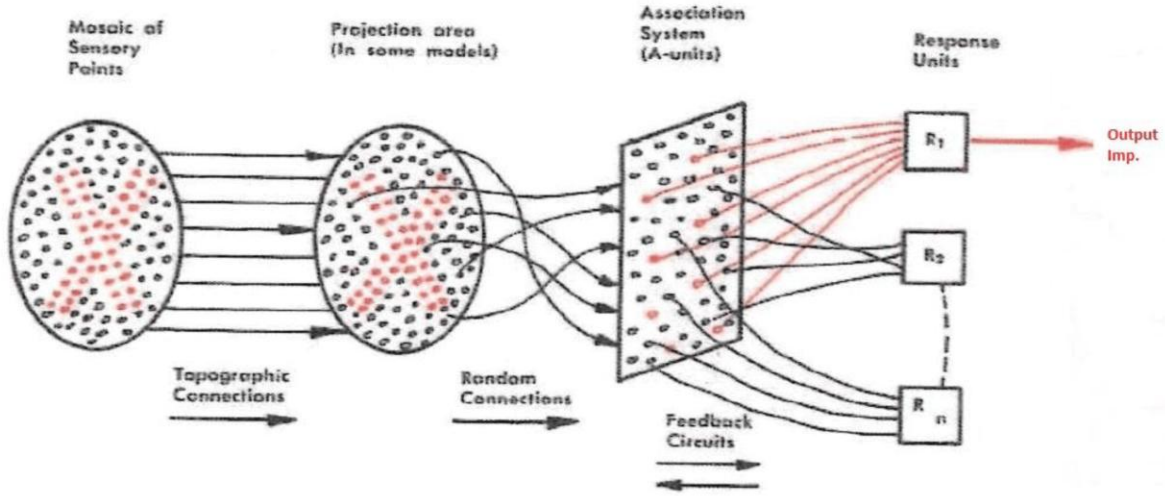


FIG. 2 — Organization of a perceptron.

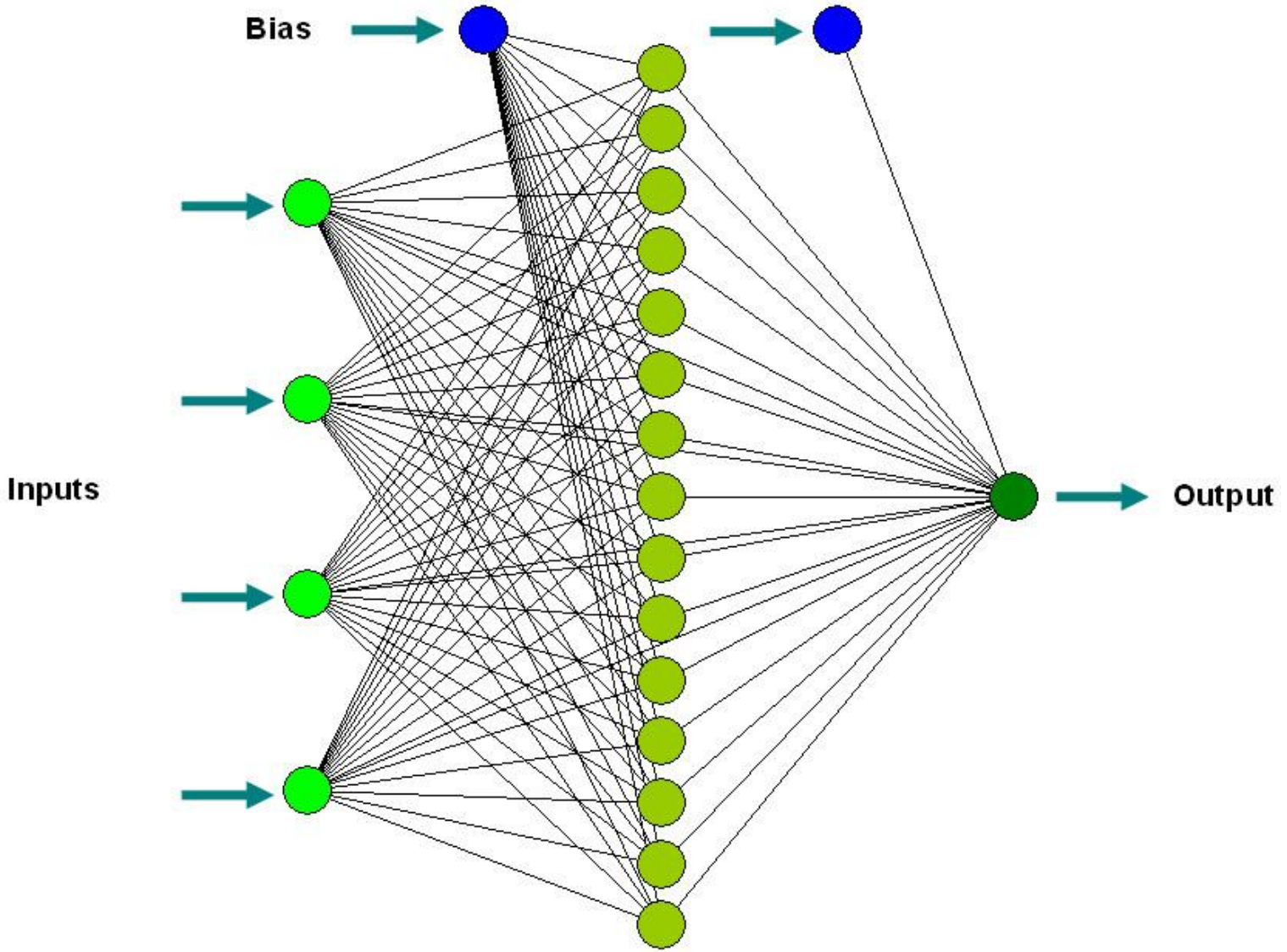
Rosenblatt, F. *The Design of an Intelligent Automaton*, Research Reviews, Office of Naval Research. Washington, October 1958, 7.

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Technology trends

Deep neural network

Basis for modern Artificial Intelligence



ÖAW AUSTRIAN ACADEMY OF SCIENCES

Technology trends



The Semantic Web

Scientific American,
May 2001
Volume 284, Issue 5

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Technology trends

- Omnipresence of the Internet
- Mobile devices
- Virtual / Augmented Reality
- Widespread use of Artificial Intelligence (machine learning, esp. deep learning)
- Reflections on the consequences of technology



INDIGO Graffiti Project, YouTube

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Artificial Intelligence

- Already being used in projects, mainly: HTR/OCR (Transkribus, Escriptoreum), NLP (spaCy)
- Many potential uses (many hitherto human tasks):
  - NLP (tagging, parsing; audio transcription; translation tasks)
  - Word Sense Disambiguation
  - AI for text encoding / annotating (editions)
  - Semantic Entity Linking
  - (Meta-)Data mining
  - Generative AI
  - Large Language Models (LLMs)
  - Vision Language Models (VLMs)
- Towards combining stochastic and knowledge-based approaches

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Training and keeping up-to-date

- Events
  - ACDH-CH Events
  - ACDH-CH Event Series (Tool Galleries, Research Days, ...)
  - University of Vienna DH Ringvorlesung (2025S)
- Open Educational Resources (OER)
  - DARIAH Campus
  - ACDH-CH How To
- Internships (at ACDH-CH) deadline **15. November 2026**

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Mailing lists

- Humanist: https://dhhumanist.org/membership_form.php
- DHd-Mailingliste: https://lists.uni-hamburg.de/mailman/listinfo/dhd
- Linguist: https://linguistlist.org/subscribe/
- Corpora: https://list.elra.info/mailman3/postorius/lists/corpora.list.elra.info/
- DH Summer Institute: https://lists.uvic.ca/mailman/listinfo/institute
- TLDR (generally on tech): https://tldr.tech
- The Batch (on AI): https://www.deeplearning.ai/the-batch/

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Thank you! Questions?

https://www.oeaw.ac.at/acdh

https://clariah.at

https://www.ditah.at

https://clarin.eu

https://dariah.eu

https://sshopencloud.eu

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